



Course Syllabus

IDEA209H · Engineering I Honors

DON BOSCO TECHNICAL INSTITUTE | TECHNOLOGY DEPARTMENT | FALL 2026

MR. PINA, apina@boscotech.edu | HOURS: 7:30, LUNCH, UNTIL 4:30

SENIOR SECTION: MR. COSSO, wcosso@boscotech.edu

Course	Full-year honors course on the daily technology block, completed in one semester, Aug 17 to Dec 11. UC A-G Area D (Science), UC honors designation.
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Enrollment	Grades 10-12. IDEA114 recommended, Integrated Math II required alongside.
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The Course

Engineering I Honors is a laboratory engineering course. You will spend most of it on your feet, at a bench or a machine, with a caliper in your hand. The order of the units is deliberate: you learn to make things before anyone asks you to design them, so your design decisions stay anchored to what the equipment can hold rather than to what looks good on a screen. Every number gets justified, every result measured, and every claim defended out loud. About 65% of class time is hands-on.

The Semester

U1	Materials and Mechanical Hardware	Identify materials by testing them. Size hardware to real loads.
U2	Manufacturing Processes	Shop certification, then five fabrication stations in rotation.
U3	Design for Manufacturability	Design it, make it, measure how close it came.
U4	Mechanism Design and Documentation	A full mechanism built from a requirement spec.
U5	Integrated Design Challenge	Four teams, four mechanisms, one working platform.

AUG 17 First day	AUG 21 Signed page due	NOV 15 Open House	DEC 11 Last class day	DEC 14 Final exam
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ideabosco.com

Everything in IDEA runs through ideabosco.com. Sign in with your boscotech.net Google account. Two parts matter every day.

IDEA CLASSROOM

ideabosco.com/classroom

Assignments are posted, opened, worked in, and submitted here, and your work saves as you go. Due dates, returned grades, comments, and the extended syllabus live here too. FACTS remains the official gradebook.

THE NOTEBOOK

ideabosco.com/notebook

You keep a physical notebook and write in it while you work, then photograph each page and upload it here. Entries attach to a specific lab or testing session, which is what makes them gradeable.

Documentation is a daily practice in this course, not something you catch up on later. **Each unit's Documentation Check is worth 25 points.**

An entry exists for every lab and testing session	7
Entries hold raw data as recorded, not reconstructed afterward	6
Entries are dated and legible	6
Entries are specific enough that another student could reconstruct your work	6

Grading

CATEGORY	WEIGHT	POINTS
Unit Labs (4 × 50)	20%	200
Unit Assignments (4 × 100)	25%	400
Engineering Documentation (4 × 25)	10%	100
Integrated Design Challenge (team 150 + individual 100)	25%	250
Final Exam	20%	100
Semester total	100%	1,050

Rubrics are checklists, and large deliverables are staged into checkpoints so an error costs you the day it happens. The final exam is an individual written problem set, Monday December 14. Letter grades and GPA weighting follow the school's scale. **Extra credit** is capped at 2% of semester points, about 21 points, purchased with IDEA Coins. That is the only extra credit in this course.

Honors designation. UC adds one grade point per semester of an approved honors course, so an A counts as 5 instead of 4, capped at 8 semesters across grades 10 and 11. You may drop the honors designation and take this as a standard 4-point course: same class, same seat, same work, only the transcript changes. Talk to your counselor first.

Three Rules That Matter

- 1 Deadlines are firm.** No extensions. An excused absence moves your due date by the school days you were out. Late work is scored on what you submitted, then reduced 10% within 5 school days and 20% after that, until the unit's grades are returned. Nothing is accepted after that, and checkpoints are excluded because they gate the next stage. There is no assigned homework, but unfinished classwork gets finished outside of class.
- 2 Defend it or fix it.** Every assignment ends with you presenting and defending your work. A number or decision you cannot defend is corrected the same day, before it earns credit. That is a correction, not a penalty.
- 3 Use AI honestly.** AI is allowed. Verify anything it gives you against the real source, and be able to explain everything you submit in your own words. Every assignment carries a level from OFF to OPEN and asks you to disclose your use. Most consumer AI services require account holders to be 18 or older, so check before you sign up for anything.

Shop Safety

- Eye protection any time anyone nearby is cutting, grinding, or machining.
- Closed-toe shoes. No loose clothing or jewelry. Long hair tied back.
- No machine runs until you are trained on it and cleared for that session.
- Hearing protection where it is called for. Bluetooth headphones and earbuds do not count.
- Report every injury, near miss, and broken tool immediately, your own included. Reporting one is never a problem. Hiding one is.
- Horseplay is removal from the shop for the day. No warning.

Accommodations. If you have a documented accommodation, bring it to me in the first two weeks and it will be honored in full. Counseling can help you start that process.

Materials

Bring every day: a notebook (purpose-made engineering notebook preferred, any bound notebook works), a pen or pencil, a calculator, and a device that reaches ideabosco.com.

Required purchase: 3M 7500 Series half facepiece respirator with 3M 60926 cartridges.

About \$45 to \$65 together, and the one item you must own to work in the shop regularly. The facepiece comes in three sizes: 7501 small, 7502 medium, 7503 large. Buy the size that fits your face. We teach you the user seal check in class before you use it, because a respirator that does not seal does not protect you. Loaners exist for the day you forget, and they are shared.

The 60926 is a multi-gas, vapor, and P100 combination cartridge, and it is the exact part number to buy. The pair is specced to one standard: a single purchase covering every operation in the IDEA shop. P100 handles metal fume from laser welding on the MetalFab plus particulate from grinding, sanding, and machining. The gas and vapor side handles formaldehyde from lasering MDF and plywood, organic vapors from adhesives and solvents, and acid gases off cut plastics. A disposable dust mask covers none of the vapor hazards.



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Recommended purchases. IDEA provides basic safety gear and required tools at all times, so nothing below is mandatory. Owning your own is recommended for fit and sanitation.

Scan for the digital syllabus, including direct links to the exact respirator and cartridges above and to every item below. If cost is a problem, see me.

Safety glasses, ANSI Z87.1+	\$8-20	Fine point markers	\$8-15
Hearing protection, NRR 25+	\$10-20	Shop apron or lab coat	\$15-25
Digital caliper, 6 in	\$15-25	USB flash drive, 64 GB+	\$8-12
Engineering notebook	\$8-12	Whole list, together	\$70-130



Syllabus Acknowledgment

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DETACH AND RETURN | DUE FRIDAY, AUGUST 21, 2026

I have read the IDEA209H syllabus. I understand:

- How the course is graded, and that extra credit is capped at 2% of semester points.
- That deadlines are firm, no extensions are given, and late work carries a 10% or 20% reduction.
- That a number or decision I cannot defend when challenged is corrected the same day before it earns credit.
- That AI use is allowed, must be verified and explainable, and must be disclosed on every assignment.
- The shop safety rules, and that violating them removes me from the shop.
- That a 3M 7500 Series respirator with 60926 cartridges is a required purchase for regular shop work, and that all other gear is provided but recommended to own.

Student

NAME (PRINT)

SECTION / BLOCK

SIGNATURE

DATE

Parent or Guardian

Students 18 or older may sign both sections themselves.

NAME (PRINT)

SIGNATURE

EMAIL

PHONE

DATE